AgriIntel — Technical Dossier 8: API Dependencies, Pricing Models & Annual Naira Cost Analysis

Exhaustive Module-by-Module Dependency Audit & FX Conversion ($1.00 USD = 1,400 NGN)

Target Architecture: AgriIntel v2.6 Core	**FX Benchmark:** $1.00 USD = NGN 1,400	**Analysis Horizon:** 1 Full Year (12 Mos)	**Baseline Cost: $0.00 / NGN 0.00**

1. Executive Summary & The Zero-Cost Architectural Principle

A foundational design objective of the AgriIntel platform is **complete scientific sovereignty and zero mandatory subscription dependency**. Unlike conventional commercial AI applications that fail when an API credit lapses or an external server throttles traffic, AgriIntel is architected with an autonomous, offline-first local biometrics engine. All mathematical computations—including Henderson's Mixed Model Equations (BLUP), Duncan's Multiple Range Test (DMRT), ANOVA, Smith-Hazel selection index matrices, stoichiometric soil balancing, and PRISMA systematic flow generation—execute deterministically on the local CPU with **zero API keys and zero monetary cost ($0.00 / NGN 0.00)**.

External APIs are utilized strictly as **optional live augmentations**. These augmentations fall into two distinct classes: (1) *100% Free & Open Scientific APIs* (Open-Meteo Weather, NASA POWER, ISRIC SoilGrids, Crossref, NCBI PubMed, FAOSTAT, CGIAR CGSpace) which require zero subscription fees; and (2) *Commercial Cloud LLM APIs* (Google Gemini, OpenAI, Groq) which provide open-ended natural language synthesis. Even within Cloud LLMs, Google Gemini offers a generous perpetual free tier (15 requests/minute), allowing academic institutions to run live AI synthesis at zero cost.

Architectural Guarantee: AgriIntel requires **NGN 0.00 ($0.00)** to install, test, and operate in production. Connecting commercial paid APIs is purely discretionary for institutions seeking high-frequency cloud LLM token generation or commercial weather SLAs. This dossier provides exact unit prices, volume models, and full annual cost projections in Nigerian Naira (NGN) at the official benchmark rate of **$1 = NGN 1,400**.

Platform API Taxonomy & Service Categorization

AgriIntel categorizes external connections into five functional technological pillars:

Category Pillar	Core Function & Usage in AgriIntel	Primary Endpoints / Providers	Default Cost Status
1. Cloud GenAI & Reasoning	Live multi-agent thesis tutoring, proposal drafting, and unstructured inquiry	Google Gemini API, OpenAI API (GPT-4o), Groq Cloud	**Freemium** (Free tier available; paid pay-as-you-go thereafter)
2. Agro-Climatology & Weather	Real-time temperature, humidity, precipitation, and livestock THI stress calculation	Open-Meteo Weather API, NASA POWER, OpenWeatherMap	**100% Free** (Open-Meteo 10k req/day free; NASA public domain)
3. Soil & Spatial Mapping	Automated edaphic diagnostics (pH, nitrogen, organic carbon, CEC) by latitude/longitude	ISRIC SoilGrids REST API (Wageningen)	**100% Free Open Data** (Creative Commons 4.0 public research)
4. Scientific Literature	Peer-reviewed evidence verification, DOI retrieval, PRISMA search protocols	Crossref REST API, NCBI PubMed E-Utilities, CGIAR CGSpace	**100% Free Open Access** (Zero subscription fees)
5. Macro Agricultural Stats	National crop yields, trade balances, fertilizer consumption, and livestock censuses	United Nations FAOSTAT REST API & Data Service	**100% Free Open Data** (UN open statistical repository)

2. Master Module-by-Module API Dependency & Connectivity Audit

The following matrix maps all 14 core modules and the Multi-Agent Orchestrator to their underlying computational engines, identifying their baseline offline capabilities versus connected external APIs, access tiers, and cost implications.

Module # & Title	Core Scientific Function	Default Engine (Always Free)	Connected External APIs	API Cost Status	Annual Cost ($ / NGN)
Mod 0: Multi-Agent Orchestrator	Dynamic multi-agent reasoning (Climate, Crop, Genetics, Soil, Econ, Evidence)	Deterministic African agricultural knowledge synthesis (13 commodities)	• Google Gemini API • OpenAI API (GPT-4o) • Groq Cloud (Llama 3.3)	Freemium (Gemini) Paid (OpenAI) Freemium (Groq)	**$0.00 / NGN 0** (Free Tier) Or ~$9.90 / NGN 13,860/yr
Mod 1: AgriResearch Copilot	Empirical Q&A, evidence search, custom file uploader, radio WAV audio	Local JSON database + in-memory Python 3-tone WAV synthesizer	• Crossref REST API • NCBI PubMed API • Google Gemini (opt)	100% Free (Crossref) 100% Free (PubMed) Freemium (Gemini)	**$0.00 / NGN 0** (All open data endpoints)
Mod 2: AI Research Scientist	Proposal builder, statistical power calculation (N), LMM equations	Local mathematical power models, RCBD experimental design templates	• Google Gemini API (opt) • OpenAI API (opt)	Freemium / Paid (Optional for extended prose)	**$0.00 / NGN 0** (Local equations default)
Mod 3: Statistical Geneticist	ANOVA, Duncan's MRT, Henderson's BLUP, breeding values (EBV)	Local SciPy, Statsmodels, Henderson MME solver. **Zero external APIs**	**None (100% Standalone)**	100% Local / Free	**$0.00 / NGN 0.00** (Runs on CPU biometrics)
Mod 4: Plant Breeding Intelligence	Smith-Hazel selection index, stay-green drought scores, radar biplots	Local multi-trait index algorithms, covariance matrix inversion	**None (100% Standalone)**	100% Local / Free	**$0.00 / NGN 0.00** (Pure local compute)
Mod 5: Soil Intelligence Engine	Edaphic health diagnostics, NPK fertilizer balancing, lime requirement	Local profile database (sample_soil_profiles_africa.csv) + chemistry logic	• ISRIC SoilGrids REST API (spatial edaphic queries)	100% Free Open Data (Creative Commons 4.0)	**$0.00 / NGN 0.00** (Zero key required)
Mod 6: Climate × Agriculture	4-scenario climate simulation, SPEI drought indices, live weather, THI	Local agro-climatic simulation matrices across 7 African climatic zones	• Open-Meteo Weather API • NASA POWER API • OpenWeatherMap (opt)	100% Free (Open-Meteo) 100% Free (NASA) Freemium (OpenWeather)	**$0.00 / NGN 0.00** (10k free daily calls on Open-Meteo)
Mod 7: Livestock Intelligence	Bovine breeding index, Avian THI math simulator, Festive broiler economics	Local multi-trait economic weights, bioclimatic differential equations	**None (100% Standalone)**	100% Local / Free	**$0.00 / NGN 0.00** (Deterministic equations)
Mod 8: Experiment Designer	Experimental layouts (RCBD, CRD, Split-Plot, Factorial), field randomization, R scripts	Local combinatorial permutation algorithms, automatic CSV/R exporter	**None (100% Standalone)**	100% Local / Free	**$0.00 / NGN 0.00** (Local procedural logic)
Mod 9: Literature Review & PRISMA	PRISMA 2020 flow diagrams, evidence matrices, gap detector, search compiler	Preloaded systematic reviews in African agro-science + local parser	• Crossref REST API • NCBI PubMed API • CGIAR CGSpace API	100% Free Open Access (All public bibliographic)	**$0.00 / NGN 0.00** (Zero subscription cost)
Mod 10: AI Thesis & Student Tutor	16-department research guidance, 10-stage thesis roadmap, Socratic viva defense	Deterministic pedagogical roadmaps, viva defense logic for 16 departments	• Google Gemini API (opt) • OpenAI API (opt)	Freemium / Paid (Optional live Socratic chat)	**$0.00 / NGN 0.00** (Free Tier covers needs)
Mod 11: Policy Simulator	Fertilizer subsidy elasticity, strategic grain reserves, tariff welfare models	Partial equilibrium microeconomic solver + multi-currency tables (5 nations)	• FAOSTAT REST API (macro agricultural statistics)	100% Free Open Data (UN FAO repository)	**$0.00 / NGN 0.00** (Open statistical data)
Mod 13: Knowledge Graph	Interactive WebGL graph mapping Gene → Trait → Soil → Climate → Market	Local NetworkX relational graph compute and Plotly WebGL renderer	**None (100% Standalone)**	100% Local / Free	**$0.00 / NGN 0.00** (Local topology engine)
Mod 15: API Hub & Diagnostics	Central management console for API keys, subscription links, and billing diagnostics	Local configuration manager within Streamlit session state	Integrates with all listed external endpoints	Administrative Console	**$0.00 / NGN 0.00** (Built-in interface)

3. Directory of Connected External APIs: Limits, Tiers & Unit Pricing

For institutions configuring live external endpoints, the table below provides granular unit costs, free quota ceilings, and direct subscription channels across all five technological domains.

API Name & Provider	Category & Domain	Free Tier Quotas & Limits	Commercial / Unit Pricing	Annual Cost ($ / NGN @ 1,400)
Google Gemini API (Google DeepMind)	GenAI Multi-Agent Reasoning `ai.google.dev`	**15 Requests/Min (RPM) Free** 1,500 req/day; 1M TPM on Flash	Flash: $0.075 / 1M in, $0.30 / 1M out Pro: $1.25 / 1M in, $5.00 / 1M out	Free Tier: **$0 / NGN 0** Paid (5k/mo): **$9.90 / NGN 13,860**
OpenAI API (OpenAI Inc.)	Alternative LLM Engine `platform.openai.com`	No free tier. Requires prepaid deposit (minimum $5.00 deposit)	GPT-4o-mini: $0.15 / 1M in, $0.60 / 1M out GPT-4o: $2.50 / 1M in, $10.00 / 1M out	Paid (5k/mo): **$19.80 / NGN 27,720** Paid (50k/mo): **$198 / NGN 277,200**
Groq Cloud API (Groq Inc.)	Low-Latency LLM Inference `console.groq.com`	**30 Requests/Min Free** 14,400 req/day on Llama 3.3 70B	On-Demand: $0.59 / 1M in, $0.79 / 1M out for enterprise throughput	Free Tier: **$0 / NGN 0** On-demand: ~$30 / NGN 42,000
Open-Meteo Weather (Open-Meteo GmbH)	Real-Time Agro-Meteorology `open-meteo.com`	**10,000 Calls/Day Free** No API key required for academic use	Commercial Tier: €29/month (~$32/mo) High-frequency business SLA	Free Tier: **$0 / NGN 0** Commercial: **$384 / NGN 537,600**
NASA POWER API (NASA Langley)	Solar Radiation & Climatology `power.larc.nasa.gov`	**100% Free & Unlimited** Public Domain US Federal science	Completely free public service. No commercial tier needed	**$0.00 / NGN 0.00** (Zero key required)
OpenWeatherMap (OpenWeather Ltd)	Commercial Weather Alt `openweathermap.org`	**1,000 Calls/Day Free** (One Call API 3.0)	$0.0015 per call beyond the first 1,000 daily free calls	Free Tier: **$0 / NGN 0** High volume: Pay-as-you-go
ISRIC SoilGrids API (ISRIC Wageningen)	Global Soil Spatial Mapping `rest.isric.org`	**100% Free Open Data** Creative Commons 4.0 license	Open access public research infrastructure. No billing system	**$0.00 / NGN 0.00** (Open REST endpoints)
Crossref REST API (Crossref)	Bibliographic Metadata & DOIs `api.crossref.org`	**100% Free Open Service** Polite pool via email header	Free non-profit academic infrastructure. No paid subscriptions	**$0.00 / NGN 0.00** (Zero key required)
NCBI PubMed API (NIH / NLM)	Biomedical & Genomic Literature `eutils.ncbi.nlm.nih.gov`	**3 req/sec open; 10 req/sec** with free NCBI user account key	Free public US Government scientific repository	**$0.00 / NGN 0.00** (Free developer account)
CGIAR CGSpace (IITA / ILRI / CIMMYT)	African Crop & Livestock Research `cgspace.cgiar.org`	**100% Free Open Access** DSpace open REST repository	Public goods agricultural research. Zero fees	**$0.00 / NGN 0.00** (Open agricultural data)
FAOSTAT Data API (UN FAO)	National Agricultural Statistics `fenixservices.fao.org`	**100% Free Open Data** Global trade, yields, livestock counts	United Nations Open Data initiative. No paid tier	**$0.00 / NGN 0.00** (Public UN service)

4. Annual Operational Tiers & Comprehensive Naira Valuation Analysis

To provide precise financial visibility for Nigerian universities, research institutes, and agribusinesses, the annual costs are evaluated across four standardized deployment scales under the exchange rate benchmark of **$1.00 USD = 1,400 NGN**.

Deployment Tier	Target Scale & User Volume	API Composition & Monthly Workload	Annual Cost (USD)	Annual Cost (NGN @ 1,400)	Effective Monthly Cost
Tier 0: Pure Academic / Starter	Individual researchers, postgraduate students, low-resource field trials	• Local deterministic biometric engines • Gemini Flash Free Tier (15 RPM) • Open-Meteo & NASA Free Tiers • Crossref, PubMed, SoilGrids Free	**$0.00**	**NGN 0.00**	**NGN 0.00 / month** (Perpetually Free)
Tier 1: Research Lab / Consulting Team	Departmental lab, 10–25 researchers, 5,000 queries/mo (~60k queries/year)	• 60M total tokens/yr (36M in, 24M out) • **Option A (Gemini 1.5 Flash)**: $9.90/yr • **Option B (GPT-4o-mini)**: $19.80/yr • Open weather, soil, literature endpoints	**$9.90** (Gemini) *$19.80 (OpenAI)*	**NGN 13,860 (Gemini)** ***NGN 27,720 (OpenAI)***	**NGN 1,155 / mo** (Gemini) *NGN 2,310 / mo (OpenAI)*
Tier 2: University Faculty / Institute	Full agricultural faculty, 500 active students & faculty, 50k queries/mo	• 600M tokens/yr (360M in, 240M out) • Gemini Flash: $99.00/yr • GPT-4o-mini: $198.00/yr • Optional Commercial Weather: $384/yr	**$99.00** (Standard) *$483.00 (with SLA)*	**NGN 138,600 (Standard)** ***NGN 676,200 (with SLA)***	**NGN 11,550 / mo** (Std) *NGN 56,350 / mo (with SLA)*
Tier 3: National Commercial Enterprise	50,000 farmers, extension agents, national advisory, 500k queries/mo	• 6.0 Billion tokens/yr (Gemini Flash: $990) • Open-Meteo Enterprise Weather SLA: $1,300 • Dedicated high-availability endpoints	**$2,290.00 / yr**	**NGN 3,206,000 / yr**	**NGN 267,167 / month** (~$0.05 / farmer / yr)

5. Detailed Mathematical Derivations & Unit Token Economics

To ensure total transparency, this section details the exact token and request volume formulas used to derive the annual budget figures.

A. Mathematical Formulation for Cloud LLM Token Consumption

For an agricultural query processed by the Multi-Agent Orchestrator, the typical prompt package includes system instructions, agro-ecological context, and user inquiries, averaging **600 input tokens**. The resulting multi-agent synthesis averages **400 output tokens**, yielding an aggregate workload of **1,000 tokens per query**.

Annual Cost Equation:

$$Cost_{annual} = [(Q_{annual} \times 600 \times P_{input}) / 1{,}000{,}000] + [(Q_{annual} \times 400 \times P_{output}) / 1{,}000{,}000]$$

Where Q_{annual} is annual query count, P_{input} is price per 1M input tokens, and P_{output} is price per 1M output tokens.

Model Engine	Query Volume (Year)	Input Tokens Cost ($)	Output Tokens Cost ($)	Annual Cost (USD)	Annual Cost (NGN @ 1,400)
Google Gemini 1.5 Flash (Lab Scale: 5k/mo)	60,000 queries (60M tokens)	36M × $0.075/1M = **$2.70**	24M × $0.30/1M = **$7.20**	**$9.90**	**NGN 13,860.00**
OpenAI GPT-4o-mini (Lab Scale: 5k/mo)	60,000 queries (60M tokens)	36M × $0.15/1M = **$5.40**	24M × $0.60/1M = **$14.40**	**$19.80**	**NGN 27,720.00**
Google Gemini 1.5 Flash (Faculty Scale: 50k/mo)	600,000 queries (600M tokens)	360M × $0.075/1M = **$27.00**	240M × $0.30/1M = **$72.00**	**$99.00**	**NGN 138,600.00**
OpenAI GPT-4o-mini (Faculty Scale: 50k/mo)	600,000 queries (600M tokens)	360M × $0.15/1M = **$54.00**	240M × $0.60/1M = **$144.00**	**$198.00**	**NGN 277,200.00**
Google Gemini 1.5 Pro (Advanced Reasoning: 50k/mo)	600,000 queries (600M tokens)	360M × $1.25/1M = **$450.00**	240M × $5.00/1M = **$1,200.00**	**$1,650.00**	**NGN 2,310,000.00**
Local Deterministic Engine (Offline Baseline: Any Vol)	Unlimited queries (Local CPU compute)	$0.00 (Local)	$0.00 (Local)	**$0.00**	**NGN 0.00 (100% Free Baseline)**

B. Comparative Analysis: Why Gemini 1.5 Flash is the Ideal Engine for Africa

1. **15 RPM Free Tier:** Google Gemini provides 15 requests per minute completely free of charge. In a typical academic seminar or field station trial, intermittent queries fall comfortably below this threshold, resulting in **an effective annual cost of NGN 0.00**.
2. **50% Lower Token Cost than GPT-4o-mini:** When moving into paid pay-as-you-go tiers, Gemini 1.5 Flash costs **$0.075 / 1M input tokens** versus $0.15 for OpenAI GPT-4o-mini, cutting operating expenditure in half.
3. **1 Million Token Context Window:** Enables ingestion of full African PhD dissertations, multi-year agro-climate time-series, or entire institutional repositories without context truncation.

6. Strategic Budget Recommendations for African Agricultural Institutions

Recommendation 1 (Zero-Budget Universities & Colleges of Agriculture):
Deploy AgriIntel in **Tier 0 Mode**. Utilize the built-in deterministic statistical geneticist (BLUP/DMRT), local soil calculators, and public Open-Meteo weather. Configure a single free Google Gemini API key via `aistudio.google.com` to unlock live LLM reasoning within the 15 RPM free ceiling. **Total Annual Budget = NGN 0.00**.

Recommendation 2 (Funded Research Institutes & CGIAR Centers - IITA/ILRI/NIHORT):
Deploy AgriIntel in **Tier 1/2 Mode**. Allocate an annual API allowance of **NGN 138,600 ($99.00 USD)** on Google Gemini 1.5 Flash. This comfortably supports up to 600,000 comprehensive multi-agent scientific analyses per year across 500 postgraduate researchers.

Recommendation 3 (Commercial Agribusinesses & Federal Ministries):
Deploy in **Tier 3 Commercial Mode**. Pair high-throughput Gemini Flash reasoning with a dedicated Open-Meteo Commercial SLA (€29–€99/mo) for continuous weather forecasting across 50,000 farmers. **Total Annual Budget = NGN 3,206,000 (~NGN 267k/month)**, representing an exceptional unit cost of approximately **NGN 64 per farmer per year**.